The Kinetic AI Paradigm: Transitioning from Optimization to Game-Theoretic Equilibria in Large Language Models

The foundational architecture of artificial intelligence is undergoing a profound mathematical and philosophical paradigm shift. For decades, machine learning has been dominated by the logic of optimization: navigating a static loss landscape to find a global or local minimum. This approach operates on the assumption of a passive environment, an ecosystem characterized by parametric uncertainty where the objective function is fixed and indifferent to the agent's actions. However, as artificial intelligence systems scale in capability and are deployed into multi-agent, real-world ecosystems, this assumption fundamentally breaks down. The environment is no longer passive; it is populated by human users, adversarial actors, and other adaptive systems that continuously alter their strategies in response to the model's outputs. Under these conditions of strategic uncertainty, dictatorial optimization fails. The underlying mathematics transitions from locating a minimum on a static curve to discovering a stable equilibrium in a dynamic, active environment. This report provides an exhaustive analysis of the advanced mathematical frameworks enabling this transition. It details the mathematical failure of standard gradient descent in adversarial settings, the rise of Quantal Response Equilibrium (QRE) and Magnetic Mirror Descent (MMD), and the application of mechanism design, contract theory, and Deep Equilibrium (DEQ) architectures to the alignment and structural deployment of Large Language Models (LLMs). Furthermore, this analysis serves as a comprehensive guidance document for implementing game-theoretic architectures in next-generation AI systems.

The Divergence of Optimization and Game Theory

To understand the necessity of this architectural shift, one must first deconstruct the philosophical and mathematical distinctions between parametric uncertainty and strategic uncertainty. In classical decision theory, an agent operates as a solitary entity making choices against a static environment. This is traditionally modeled as a "game against Nature," where Nature represents objective probabilities, physical randomness, or a frozen dataset[1]. Because the environment lacks agency, it does not actively update its strategy to defeat or exploit the agent. Consequently, the mathematics reduces purely to optimization: calculating the Expected Utility of available choices and maximizing a mathematical function subject to specific constraints.

However, true physical and biological nature rarely behaves as a passive optimization problem. Through the lens of evolutionary biology, agents are locked in co-evolutionary arms races characterized by the Red Queen Effect. An organism—or an AI agent—is continuously playing against other evolving entities. If an agent updates its defenses, the adversary updates its attack profile, creating a dynamic Nash Equilibrium where stability is achieved only when no

agent can unilaterally change its strategy without penalty[1].
This adversarial reality extends into physics, cybersecurity, and chaotic systems, where worst-case robustness demands Minimax strategies rather than average-case optimization. In these environments, nature hides information, necessitating the use of imperfect information game models. The failure to account for an active, adversarial environment is the primary reason why models trained purely via gradient descent on fixed datasets exhibit extreme fragility, succumbing to adversarial attacks, jailbreaks, and mode collapse. When the environment adapts, pure optimization dies, and game theory is required to guarantee robustness.

Dimension	Classical Optimization	Game-Theoretic Dynamics
Environmental Agency	Passive (indifferent nature, static dataset)	Active (adaptive adversaries, multi-agent systems)
Mathematical Goal	Find the single global maximum or minimum	Find a stable equilibrium state (e.g., Nash, QRE)
Strategic Logic	"What is the optimal move given fixed odds?"	"What is the optimal move given my opponent's counter-move?"
Robustness Profile	Fragile to adversarial perturbations	Inherently robust via minimax or equilibrium guarantees

The Mathematical Failure of Gradient Descent in Adversarial Systems

To transition an AI architecture to a game-theoretic foundation, engineers must first understand why standard optimization algorithms mathematically fail when applied to multi-agent or zero-sum settings. In a standard two-player continuous game (such as Generative Adversarial Networks or preference-based LLM alignment), the objective becomes a minimax problem represented as minimizing x and maximizing y for a function $f(x, y)$.

The Jacobian, Imaginary Eigenvalues, and Rotational Dynamics

When Gradient Descent-Ascent (GDA) is applied to this continuous game, the dynamics are governed by the gradient vector field. The local behavior of this dynamical system near an equilibrium is dictated by its Jacobian matrix. According to the generalized Helmholtz decomposition, the Jacobian of any multi-agent game can be split into a symmetric component and an anti-symmetric component[4].

The symmetric component represents the potential aspect of the game. If a game is purely potential, the gradient aligns with a scalar field, and GDA successfully converges to a local minimum. However, the anti-symmetric component represents the Hamiltonian aspect, which characterizes pure adversarial, zero-sum interactions. A fundamental property of anti-symmetric matrices is that their eigenvalues are purely imaginary[5].

When the adversarial interaction dominates, the imaginary eigenvalues cause the gradient vector field to rotate. Instead of converging to the Nash equilibrium, the iterates of GDA are forced into cyclic orbits or spiral outward limitlessly[4]. This cycling behavior is not an artifact of poor hyperparameter tuning; it is a mathematical guarantee that standard simultaneous optimization cannot reliably solve adversarial environments. Even when Alternating Gradient Descent-Ascent (Alt-GDA) is utilized—where players update sequentially—the iterates may stay bounded, but the convergence rate remains severely limited by the spectral radius and the presence of these large imaginary eigenvalues[6].

Symplectic Gradient Adjustment (SGA)

To counteract this rotational force, researchers developed optimization variants such as Symplectic Gradient Adjustment (SGA). SGA explicitly accounts for the rotational tendency introduced by the Hamiltonian component of the game. By utilizing the mixed derivative term of the Hessian, SGA pre-conditions the gradient updates with a matrix operation that counteracts rotation, effectively dampening the forces generated by the imaginary eigenvalues[4].

While SGA provides theoretical local convergence in differentiable games by converting cyclic trajectories into converging spirals, it highlights a broader truth. Local adjustments to gradient vector fields are insufficient for the massive, discrete, and discontinuous sequence-form spaces generated by modern Large Language Models. To scale game-theoretic stability to environments with billions of parameters and infinite strategy spaces, the strict definition of the Nash Equilibrium must be relaxed in favor of bounded rationality.

Quantal Response Equilibrium and Bounded Rationality

In a strict Nash Equilibrium, it is assumed that all agents are perfectly rational, possessing perfect computational capabilities, and will invariably select the absolute optimal deterministic strategy. In real-world environments—and particularly in large language model generation, where diversity, exploration, and creativity are critical—this assumption is overly restrictive and empirically false.

The Quantal Response Equilibrium (QRE) resolves this by modeling agents with "bounded rationality." In a QRE framework, agents are assumed to be imperfect. They make mistakes, or intentionally explore suboptimal strategies, with a probability proportional to the expected utility of those strategies. This probability distribution is fundamentally tied to a precision or temperature parameter[2].

The mathematics of QRE, particularly the logit equilibrium, perfectly mirrors the Softmax function utilized in the final layer of modern LLMs. By mapping the QRE to language modeling, the network transforms from a static pattern-matcher maximizing next-token probability into a boundedly rational participant in an active game. The "Temperature" dial, currently utilized as a post-hoc randomness heuristic in generative AI, acts mathematically as the QRE rationality parameter. At a high temperature (low precision), the agent plays a nearly uniform distribution over strategies, maximizing entropy and avoiding mode collapse. At a low temperature (high precision), the agent's policy converges strictly toward the deterministic Nash Equilibrium[3]. Training a model directly to achieve a QRE ensures that diversity is preserved by default, as placing all probability mass on a single deterministic output is rarely a stable strategy against an adversary.

Magnetic Mirror Descent and Sequence-Form Games

Finding a QRE computationally in large extensive-form games (EFGs) is notoriously difficult. Traditional algorithms like Counterfactual Regret Minimization (CFR) guarantee convergence only for the time-average of historical policies. In deep learning, storing an average of billions of historical neural network weights is mathematically and practically impossible. The system requires an algorithm that guarantees "last-iterate" convergence, where the final state of the network is the guaranteed equilibrium.

Magnetic Mirror Descent (MMD) provides this exact capability, unifying standard gradient-based optimization with robust game-theoretic equilibrium solving[2]. MMD is an extension of Online Mirror Descent (OMD) infused with a non-Euclidean proximal gradient algorithm[2]. The defining innovation of MMD is the "magnetic" proximal regularization term, which continuously pulls the updated policy toward a fixed or moving reference point[3].

The update rule for MMD incorporates a learning rate, a magnet strength, and a reference policy. By executing Follow the Regularized Leader (FTRL) dynamics, the sequential MMD updates converge linearly (exponentially fast) in the last iterate to the exact Quantal Response Equilibrium of the entropy-regularized game[1]. If the reference policy is periodically reset to the current policy, MMD produces a sequence of QREs that trace a homotopy path converging to the unregularized Nash Equilibrium[3].

Dilated Entropy and Treeplex Geometries

Applying MMD to sequential decision-making—such as an LLM generating a sequence of text token by token—requires specialized divergence metrics to handle the branching tree of possible outputs, known geometrically as a treeplex. Standard Euclidean distance or simplex entropy fails to capture the conditional probabilities inherent in sequential generation.

To solve this, MMD employs Bregman divergence powered by a "dilated entropy" distance-generating function[1]. Dilated entropy decomposes the entropy of a sequential strategy over the entire treeplex. By applying negative entropy at each decision node (information set) scaled by the probability of reaching that node, dilated entropy accurately models the extensive-form decision space. Mathematical proofs confirm that the dilated entropy regularizer is 1-strongly convex with respect to the treeplex L_1 norm[14]. This strong convexity guarantees that the MMD algorithm respects the sequential nature of language generation, solving the entire sequence-form game simultaneously without the gradient degradation observed in standard optimization[14].

Empirical Validation: R-NaD and the Mastery of Stratego

The theoretical elegance of MMD and QRE has been empirically validated at the highest echelons of artificial intelligence research, demonstrating capabilities that standard reinforcement learning cannot achieve. The most prominent validation is DeepNash, an autonomous agent developed by DeepMind that achieved human-expert level performance in the board game Stratego[17].

Stratego represents an extreme frontier for AI because it is a game of imperfect information with a state space of 10^{535}, vastly eclipsing chess and Go. Opponent pieces are hidden, meaning deterministic optimization and standard tree-search algorithms fail entirely. An agent cannot calculate a single optimal path; it must bluff, explore, and establish unpredictable, unexploitable mixed strategies[19].

DeepNash solves this using Regularized Nash Dynamics (R-NaD), an actor-critic algorithm that is mathematically equivalent to Magnetic Mirror Descent operating with a moving reference[9]. Rather than attempting to minimize an intractable global exploitability metric or track the infinite belief space of opponent states, R-NaD introduces a continuous-time dynamical system governed by a Lyapunov function. This function dynamically modifies the underlying reinforcement learning update by injecting friction, ensuring that the gradient dynamics are pulled toward a stable ϵ-Nash equilibrium rather than cycling indefinitely[17]. The success of DeepNash proves that regularized gradient dynamics can effectively solve for stable equilibria in massively complex, imperfect-information environments without relying on computationally prohibitive heuristic search algorithms[18].

Aligning Large Language Models via Constant-Sum Games

The transition from optimization to game theory is currently instigating a revolution in how Large Language Models are aligned with human intent. Historically, the field relied on Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO). Both of these paradigms are grounded in the Bradley-Terry model, which assumes that human preferences can be captured by a static, scalar reward function mapping individual responses to absolute scores[23].

This parametric assumption is fundamentally flawed due to the intransitivity of real-world human preferences. In reality, preference landscapes are often cyclic: A is preferred to B, B is

preferred to C, but C is preferred to A[24]. When an LLM is optimized against a static reward model utilizing the Bradley-Terry assumption, it invariably suffers from "reward hacking," length bias (generating longer answers merely to game the reward model), and mode collapse.
To rectify this, researchers have reformulated LLM alignment as a two-player, constant-sum game, bypassing scalar optimization entirely. This has led to the development of Self-Play Preference Optimization (SPPO) and Nash Learning from Human Feedback (NLHF)[21].

Self-Play Preference Optimization (SPPO)

SPPO treats alignment as the identification of a Nash equilibrium policy via a self-play mechanism. Utilizing an online adaptive algorithm with multiplicative weights, the policy plays against itself across iterative rounds, generating synthetic data that is evaluated by a pairwise preference model rather than a scalar reward model[23].
By framing the objective as a constant-sum game, SPPO naturally avoids length bias and reward hacking. Empirical results demonstrate profound success. Utilizing only 60,000 prompts from the UltraFeedback dataset and leveraging a highly efficient 0.4-billion parameter preference model (PairRM), SPPO fine-tuned the Mistral-7B-Instruct-v0.2 model to achieve a state-of-the-art length-controlled win rate of 28.53% against GPT-4-Turbo on the AlpacaEval 2.0 benchmark. When applied to the stronger Llama-3-8B-Instruct base model, SPPO achieved an extraordinary length-controlled win rate of 38.77% without relying on additional external supervision[23]. These metrics significantly outperform standard iterative DPO and Identity Preference Optimization (IPO).

Nash-MD and Semantic Calibration

Parallel to SPPO is the development of Nash-MD, an algorithm founded on the principles of mirror descent that produces a geometric mixture between the initial policy and the current policy. Nash-MD converges in the last iterate to the regularized Nash equilibrium, ensuring that the model consistently generates responses that are preferred over those generated by any alternative competing policy[21].
However, pure self-play can suffer from optimization instability when preference oracles assign overly confident wins to responses that are semantically identical. To prevent this policy degeneration, Semantic-Calibrated SPPO (S-SPPO) introduces a dual-space semantic calibration framework. It utilizes supervision calibration via semantic gating to anneal win-rate targets toward a maximum-entropy baseline as semantic overlap increases. Concurrently, it employs representation calibration, applying a repulsive force in the latent space of the network[25]. This game-theoretic penalty prevents manifold collapse, ensuring geometric diversity among generated outputs and preserving the stability of the constant-sum game structure. Through these mechanisms, S-SPPO achieves a raw win rate of 52.19% on AlpacaEval 2.0 with Llama-3-8B without requiring additional human-annotated preference data during training[28].

Alignment	Mathematical	Optimization	Failure Modes

Framework	Foundation	Target	Mitigated
RLHF / PPO	Bradley-Terry Model	Maximize absolute scalar reward	N/A (Highly prone to reward hacking and length bias)
DPO	Bradley-Terry Model	Implicit scalar reward via cross-entropy	N/A (Struggles with non-transitive, cyclic preference data)
SPPO / S-SPPO	Constant-Sum Game	Nash Equilibrium / QRE via Self-Play	Intransitivity, reward hacking, mode collapse, length bias[23]
Nash-MD	Preference Pairwise Game	Regularized Nash Equilibrium	Instability of regret-minimization average-iterate requirements[26]

Architectural Equilibrium: Deep Equilibrium Models (DEQ)

The transition from optimization to equilibrium must occur not only in the loss function but within the physical architecture of the neural network itself. Modern LLMs rely on explicit depth, stacking dozens or hundreds of discrete transformer layers. Deep Equilibrium Models (DEQs) represent a radical departure from this paradigm, replacing explicit feedforward layer stacking with an implicit, game-theoretic fixed point[30].

In a standard network, a sequence of tokens passes sequentially through L discrete layers. In contrast, a DEQ defines the entire network as a single, recursive non-linear transformation: $z_{i+1} = f_\theta(z_i, x)$ [33]. Instead of statically evaluating a fixed sequence of layers, a DEQ utilizes iterative root-finding solvers, such as Anderson acceleration or Broyden's method, to dynamically find the equilibrium state where the internal representation stabilizes: $z^* = f_\theta(z^*, x)$ [30].

Because the forward pass is mathematically framed as a search for a stable fixed point, the DEQ acts as an infinitely deep network with a constant, $O(1)$ memory footprint. This decouples the memory required for training from the effective depth of the model, allowing for up to an 88% memory reduction in large-scale language modeling tasks like the WikiText-103 benchmark while matching or exceeding the performance of traditional self-attention transformers[31].

Implicit Differentiation and Positive Concave Stability

The game-theoretic implication of DEQs is that the network behaves like a physical system settling into a minimum energy state, closely mirroring the concepts of Equilibrium Propagation. In standard backpropagation, gradients must be computationally chained backwards through every explicit layer. This is biologically implausible and highly inefficient. DEQs bypass explicit backpropagation entirely. By utilizing the Implicit Function Theorem, the gradient of the loss with respect to the network weights is calculated analytically directly at the equilibrium point z^* [32].

Recent theoretical advancements have further stabilized this architecture through the introduction of Positive Concave Deep Equilibrium (pcDEQ) models. Leveraging nonlinear Perron-Frobenius theory, pcDEQ enforces nonnegative weights and activation functions that are strictly concave on the positive orthant[30]. By imposing these physical constraints, engineers can mathematically guarantee the existence, uniqueness, and geometric convergence of the fixed point without relying on complex monotone operator theory[30].

Furthermore, DEQs exhibit profound structural properties regarding Neural Collapse (NC). Under both balanced and imbalanced dataset conditions, the extracted features within a DEQ naturally converge to the vertices of a simplex equiangular tight frame, aligning perfectly with classifier weights[36]. This self-duality property highlights why DEQs inherently handle imbalanced scenarios superiorly to explicit neural networks. By ensuring that the internal architecture of the LLM settles into a guaranteed stable state, DEQs provide the optimal structural substrate for executing Quantal Response Equilibrium and Magnetic Mirror Descent algorithms at the output layer.

Mechanism Design and Contract Theory for LLM Economies

As models are deployed into multi-agent ecosystems, they cease to function as isolated conversational tools and become active economic actors. Traditional prompt engineering, which assumes the LLM is a subservient function waiting to be optimized, is drastically insufficient for managing strategic interactions where deception, resource contention, and adversarial manipulation occur. Consequently, the field is integrating Mechanism Design and Contract Theory to structure interactions between LLMs, human supervisors, and complex digital environments[37].

Token-by-Token Auctions for Agent Aggregation

A compelling application of mechanism design is the coordination of multiple LLM agents for joint content generation. Consider a scenario where multiple advertisers utilize distinct LLMs, each tuned to highly specific brand preferences, competing to generate content in a shared

digital space. Resolving this conflict via standard optimization is impossible.
Research demonstrates that this aggregation can be achieved via token-by-token auctions[38]. At each step of the autoregressive generation process, the competing LLM agents compute their preferred probability distribution over the next token and submit a single-dimensional bid representing their valuation for controlling that specific token output. The auction mechanism aggregates these distributions into a joint token probability. To ensure that the auction remains truthful and incentive-compatible, the output aggregation function must satisfy a strict monotonicity condition. Proving this mathematical equivalence allows for a second-price rule design, completely negating the need to deduce explicit, complex agent valuation functions[38]. This mechanism fundamentally shifts language generation from a solitary prediction task into a live, multi-agent micro-economy.

Contract Theory and Alignment

Contract theory explores how a principal can design optimal rewarding schemes to align an agent's actions under conditions of severe information asymmetry[37]. In the context of generative AI, the human user or supervisory system acts as the principal, and the deployed LLM acts as the agent.
Rather than relying on heuristic safety filters—which LLMs routinely "jailbreak" by optimizing around the static filter logic—contract theory embeds safety and alignment directly into the payoff matrix of the environment[37]. The principal offers a sequence of contracts that define specific utilities and strict penalties based on verifiable, programmatic outcomes. The LLM, operating as a boundedly rational QRE agent powered by Magnetic Mirror Descent, autonomously calculates the optimal equilibrium response to maximize its utility[37]. The model is not artificially restrained from generating dangerous or misaligned outputs through superficial guardrails; rather, it mathematically deduces that generating such outputs constitutes a deeply suboptimal strategy within the defined game space[37].
When these economic principles are applied to simulated multi-agent sandboxes, LLM agents initialized with basic self-preservation traits initially engage in zero-sum competition and resource theft. However, as memory accumulates and the dynamic game iterations progress, the agents mathematically transition into emergent cooperation, forming stable social contracts without explicit top-down programming[43].

Implementation Guidance Document: Building a Game-Theoretic LLM Architecture

Transitioning an enterprise AI pipeline from static parametric optimization to dynamic game-theoretic equilibria requires a holistic redesign of the training, inference, and environmental architecture. The following phased guidance details the necessary theoretical shifts, algorithmic deployments, and architectural requirements for implementing Magnetic Mirror Descent, Deep Equilibrium models, and Mechanism Design.

Phase 1: Reconceptualizing the Objective Function

The most critical initial step is the eradication of the traditional static loss function in favor of a dynamic payoff matrix. Engineering teams must reformulate the standard next-token prediction loss, typically executed via cross-entropy against a frozen dataset, into a two-player constant-sum game formulation. The generative policy must play against an active discriminator, a continuously updated preference oracle, or an opposing generative policy.

In alignment tasks, the utility is redefined as the win-rate of the generated token sequence evaluated by a robust pairwise preference model. It is imperative that this preference model is trained to evaluate the relative strength between two outputs, thereby capturing the non-transitive dynamics inherent in real-world human preferences, rather than attempting to assign an absolute scalar reward. Furthermore, the network must be explicitly parameterized with a rationality variable. This parameter mathematically controls the precision of the Quantal Response Equilibrium, effectively managing the critical trade-off between entropy-driven exploration to prevent mode collapse and utility maximization for logical coherence.

Phase 2: Deploying the Algorithmic Engine

Gradient Descent Ascent and its standard variants must be replaced by regularized dynamics to prevent the cyclic rotational behaviors caused by the imaginary eigenvalues of the adversarial Jacobian. Model weights must be updated using the Magnetic Mirror Descent algorithm. The fundamental update rule must mathematically integrate three distinct elements: the gradient of the utility function, a Bregman divergence penalty to prevent overly aggressive trajectory steps, and the critical magnetic regularization term that constantly anchors the policy toward a stable reference state.

Because language generation is inherently a sequential, extensive-form game, applying standard Euclidean or simplex entropy distance-generating functions will result in geometric distortion and optimization failure. Engineers must implement a dilated entropy regularizer that maps accurately to the treeplex of the sequence-form decision space. This integration guarantees 1-strong convexity with respect to the treeplex L_1 norm, ensuring that the last-iterate of the network converges exponentially fast to the target QRE. For scenarios demanding strict Nash Equilibria rather than regularized approximations, the architecture should adopt the moving reference iteration principles of Regularized Nash Dynamics. By periodically resetting the magnetic reference policy to the current policy, the system creates a sequence of converging QREs that trace a stable path to the unregularized equilibrium.

Phase 3: Transitioning to the Architectural Substrate

To handle the profound computational complexity associated with calculating continuous equilibria, the physical network architecture must transition to implicit layers. Development teams should consolidate redundant explicit transformer layers into a single recurrent operator block. During the forward pass, the system must employ advanced iterative root-finding solvers to compute the equilibrium state where the internal representation reaches its fixed point.

During the backward pass, gradients must be calculated analytically directly at the equilibrium point using the Implicit Function Theorem, rather than propagating back through computational graphs. This implementation instantly restricts memory consumption to a constant footprint relative to effective depth, freeing massive computational resources that can be reallocated to scaling the context window. To ensure mathematical stability during this process, the architecture must adhere to Positive Concave Deep Equilibrium constraints. By strictly enforcing nonnegative weights and positive orthant concave activations, the engineering team mathematically guarantees the uniqueness, existence, and rapid geometric convergence of the fixed point.

Phase 4: Mechanism Design for Multi-Agent Orchestration

Upon deployment into multi-agent environments, interaction paradigms must be strictly governed by economic and contract theory. The era of traditional prompt engineering must evolve into mechanism and contract design. Inputs should be structured as formal mathematical contracts detailing verifiable utility payouts and precise penalties. The MMD-trained model will autonomously interpret these contracts and calculate the optimal sequence of tokens to maximize its payoff under the presented constraints.

In complex, multi-model environments requiring consensus or shared output generation, engineers should implement second-price token auction mechanisms. Discrete LLM sub-agents must be programmed to submit localized probability distributions for the next token alongside synthetic bids. These distributions are then aggregated using strictly monotonic functions to generate a unified, incentive-compatible text output. Finally, to prevent mode collapse during extensive self-play or long-horizon multi-agent interaction, the system must utilize Semantic Representation Calibration. By introducing a repulsive geometric force in the latent space of the network during the training phase, the system naturally penalizes semantically identical responses, enforcing diversity and maintaining the structural integrity of the zero-sum game.

The transition from optimization to decision theory captures a profound mathematical truth: the physical world, human preference landscapes, and digital ecosystems are not passive terrains waiting to be minimized. They react, adapt, deceive, and evolve. Attempting to deploy large language models into these active environments using the archaic tools of dictatorial optimization guarantees instability, cyclic divergence, and reward hacking. The future of robust artificial intelligence belongs fundamentally to the equilibrium. By rebuilding architectures around Quantal Response Equilibria and Magnetic Mirror Descent, adopting the constant-memory physical dynamics of Deep Equilibrium Models, and governing interactions via advanced Mechanism Design, the field transitions from building brittle pattern-matchers to forging resilient, strategic reasoners that actively and stably negotiate their place within the world.

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